Marks	60-63	63-66	66-69	69-72	72-75
No. of students	5	18	42	27	8

Solution:

L = Upper limit of highest class = 75

S = lower limit of lowest class = 60

Range = L-S = 75-60 = 15

Co - efficient of Range = (L- S) / (L+ S)

= 15/ (75 + 60) = 15 / 135=0.111

Merits:

- Range is the simplest measure of dispersion.
- It is well defined, and easy to compute.
- It is widely used in quality control, weather forecasting, stock market variations etc.

Limitations:

- The calculations of range is based on only two values – largest value and smallest value.
- It is largely influenced by two extreme values.
- It cannot be computed in the case of open-ended frequency distributions.
- It is not suitable for further mathematical treatment.

6.3.2 Inter Quartile Range and Quartile Deviation

The quartiles Q_1, Q_2 and Q_3 have been introduced and studied in Chapter 5.

Inter quartile range is defined as: Inter quartile Range (IQR) = $Q_3 - Q_1$

Quartile Deviation is defined as, half of the distance between Q_1 and Q_3.

Quartile Deviation Q.D = $\frac{Q_3 - Q_1}{2}$

It is also called as semi-inter quartile range.

Ch6_11th Stats.indd 160 2/3/2019 11:01:51 AM